The Grassroots-to-Mastery Roadmap

Phase 2: Spatial Vector Topology

The Exhaustive Vector Group Masterclass

A Complete Geometric and Mathematical Deconstruction of Every Standard 3-Phase Transformer Connection

Module Objective:

This extended volume leaves no stone unturned. You requested every single vector group. You will receive every single vector group. We will systematically construct the wiring diagrams, derive the phasor geometries, and plot the clock configurations for Group I (0°), Group II (180°), Group III (−30°), and Group IV (+30°), encompassing Yy, Dd, Dy, Yd, and Yz topologies.

Pedagogical Sequence Framework

Phase 2 of 6 (Extended Edition)

May 22, 2026

Contents

1 The Mathematics of Spatial Phase Shift

Cognitive Thought Process & Engineering Logic

1. Before memorizing clock numbers, we must prove mathematically why a 30° shift is geometrically inevitable when you mix Star and Delta connections.
2. The golden rule of transformers: Primary and secondary coils wound on the SAME physical iron limb must have phase voltages that are perfectly in phase (or perfectly 180° out of phase if polarities are reversed).
3. We will map a Delta primary (where line voltage = phase voltage) to a Star secondary (where line voltage = phase difference) to expose the inherent 30° offset.

1.1 The Winding Core Constraints

In any multi-phase transformer bank or three-limb core structure, the primary and secondary phase windings wrapped around the *exact same iron limb* are linked by the same mutual core flux (Φ). Consequently, the induced phase voltages across those specific concentric coils **must remain perfectly co-linear** in the time domain.

Let us define the core limb induced phase voltages as:

$$V_{A\text{-phase}} \parallel V_{a\text{-phase}}, \quad V_{B\text{-phase}} \parallel V_{b\text{-phase}}, \quad V_{C\text{-phase}} \parallel V_{c\text{-phase}} \tag{1}$$

1.2 Rigorous Proof of $\Delta - Y$ Angular Shifts

Consider a Delta-Star ($\Delta - Y$) configuration. The primary line-to-line voltages are directly bounded across the phase windings. Let the primary line voltages be:

$$\vec{V}_{AB} = V_L \angle 0^\circ \tag{2}$$
$$\vec{V}_{BC} = V_L \angle -120^\circ \tag{3}$$
$$\vec{V}_{CA} = V_L \angle +120^\circ \tag{4}$$

Because winding $A - B$ is physically mounted on Limb 1, the corresponding secondary phase winding $a - n$ on Limb 1 must mirror this identical orientation. Hence, the secondary line-to-neutral phasor $\vec{V}_{an}$ is forced to be co-linear with $\vec{V}_{AB}$:

$$\vec{V}_{an} = V_{ph,sec} \angle 0^\circ \tag{5}$$

Similarly, for the remaining limbs:

$$\vec{V}_{bn} = V_{ph,sec} \angle -120^\circ, \quad \vec{V}_{cn} = V_{ph,sec} \angle +120^\circ \tag{6}$$

Now, let us compute the secondary system's resultant **line-to-line voltage** $\vec{V}_{ab}$ using spatial phasor addition. By Kirchhoff's Voltage Law:

$$\vec{V}_{ab} = \vec{V}_{an} - \vec{V}_{bn} \tag{7}$$

Substituting our complex angular values:

$$\vec{V}_{ab} = V_{ph,sec}\angle 0^\circ - V_{ph,sec}\angle -120^\circ \tag{8}$$
$$= V_{ph,sec}\left[(1+j0) - \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right)\right] \tag{9}$$
$$= V_{ph,sec}\left[\frac{3}{2} + j\frac{\sqrt{3}}{2}\right] \tag{10}$$
$$= \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \cdot V_{ph,sec}\angle \tan^{-1}\left(\frac{\sqrt{3}/2}{3/2}\right) \tag{11}$$
$$= \sqrt{3}V_{ph,sec}\angle 30^\circ \tag{12}$$

Comparing the primary system line voltage $\vec{V}_{AB} = V_L\angle 0^\circ$ against the secondary system line voltage $\vec{V}_{ab} = \sqrt{3}V_{ph,sec}\angle 30^\circ$, we observe an unavoidable spatial displacement:

$$\angle\vec{V}_{ab} - \angle\vec{V}_{AB} = +30^\circ \text{ (or } -30^\circ \text{ depending on terminal wiring labels)} \tag{13}$$

Thus, vector group formation with an implicit 30° phase shift is structurally and mathematically **unavoidable** whenever a transformer bridges a delta configuration with a star configuration.

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2 The Universal Clock Convention

MANDATORY MULTIMEDIA INTEGRATION: YOUTUBE LECTURE

Video Topic: Transformer Clock Convention Explained
Search Query / Link: https://www.youtube.com/results?search_query=transformer+clock+convention+phasor
Required Focus: Watch how the primary phasor is locked at 12 o'clock. Observe how swapping standard $A-B-C$ sequences to $A-C-B$ causes the hour hand to jump across the clock face.

Cognitive Thought Process & Engineering Logic

1. Engineers need a universal framework to designate these complex phase shifts without writing out full equations on nameplates.
2. The standard analog clock dial offers an elegant circular coordinate mapping space ($360^\circ/12$ hours $= 30^\circ$ per hour).
3. The High Voltage (HV) line phasor is locked permanently as the reference Minute Hand at 12 o'clock (0°).
4. The Low Voltage (LV) line phasor acts as the Hour Hand, moving dynamically to indicate the phase shift.

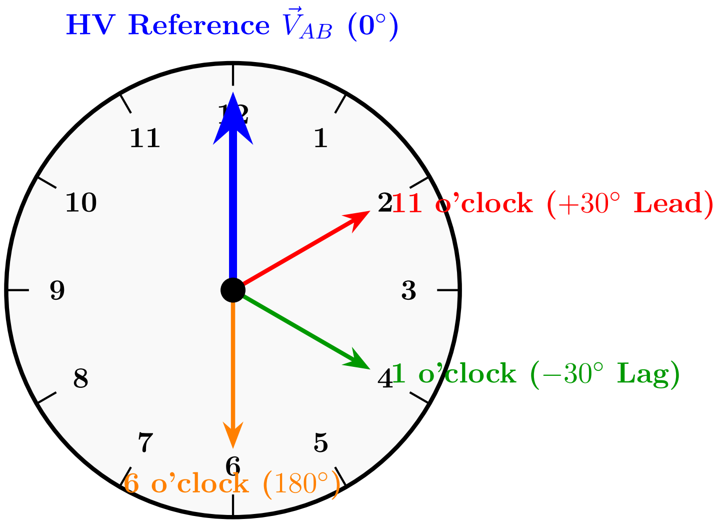

Figure 1: The Universal Clock Face Mapping. The rotation of the secondary voltage indicates its phase relation to the primary.

3 Group I: Zero Phase Displacement (0° / 12 o'clock)

Cognitive Thought Process & Engineering Logic

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Group I covers transformers where the primary and secondary line voltages
are perfectly in-phase. This occurs when both sides have the same
configuration (Star-Star, Delta-Delta) and are wired with identical
polarities. The hour hand rests directly on 12.
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3.1 1. Vector Group: Yy0

Configuration: Star Primary (Y), Star Secondary (y), 0° Shift.

- **Wiring Logic:** Primary terminals A_1, B_1, C_1 are shorted to form neutral N. Lines feed A_2, B_2, C_2. Secondary terminals a_1, b_1, c_1 are shorted to form neutral n. Outputs are taken from a_2, b_2, c_2.
- **Phasor Result:** Since windings on the same limb have the same polarity, $\vec{V}_{AN}$ is in-phase with $\vec{v}_{an}$. Consequently, the line voltages $\vec{V}_{AB}$ and $\vec{v}_{ab}$ are perfectly in phase.

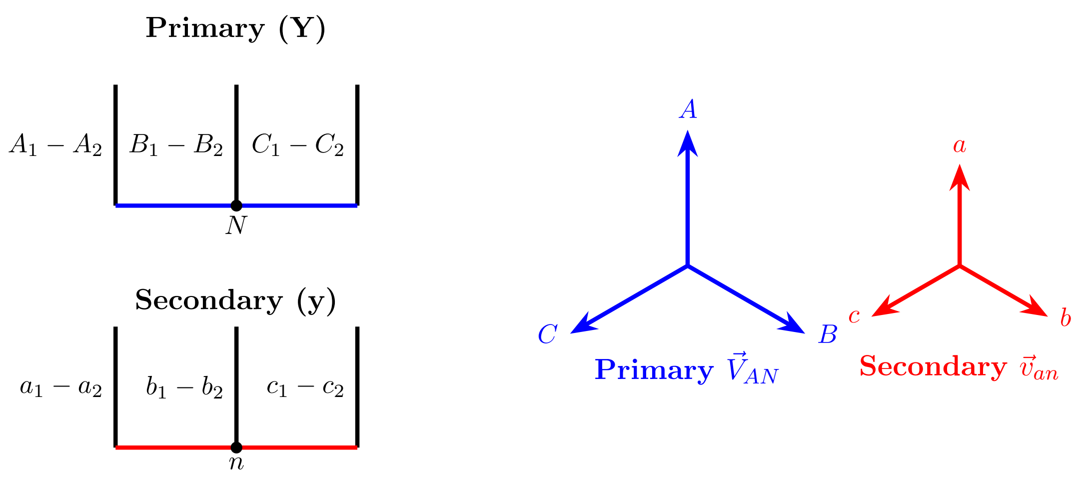

Figure 2: Yy0 Wiring and Phasor alignment. Both systems are identical, generating a 12 o'clock alignment.

3.2 2. Vector Group: Dd0

Configuration: Delta Primary (D), Delta Secondary (d), 0° Shift.

- **Wiring Logic:** Primary connects $A_2 \to B_1$, $B_2 \to C_1$, $C_2 \to A_1$. Secondary matches this exactly: $a_2 \to b_1$, $b_2 \to c_1$, $c_2 \to a_1$.
- **Phasor Result:** The primary line voltage directly equals the primary phase voltage. The secondary mirrors this, resulting in 0° shift.

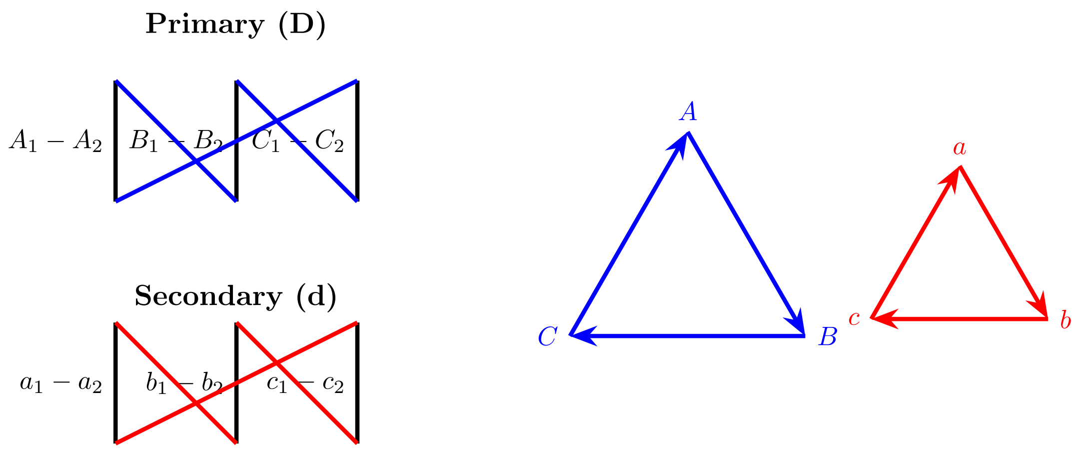

Figure 3: Dd0 Wiring and Phasor alignment. Both triangles are perfectly synchronized at 12 o'clock.

4 Group II: 180° Phase Displacement (6 o'clock)

Cognitive Thought Process & Engineering Logic

`What happens if we take a perfectly aligned 0-degree transformer (like Yy0) but we deliberately wire the secondary coils backwards? We flip the polarity of every secondary coil by 180°. The Hour Hand jumps from 12 to 6.`

4.1 1. Vector Group: Yy6

Configuration: Star Primary (Y), Star Secondary (y), 180° Shift.

- **Wiring Logic:** Primary connects A_1, B_1, C_1 to form neutral. To invert the secondary, we connect the *opposite* ends (a_2, b_2, c_2) to form the neutral. Outputs are taken from the a_1, b_1, c_1 ends.
- **Phasor Result:** Secondary phase voltages point exactly opposite to primary phase voltages.

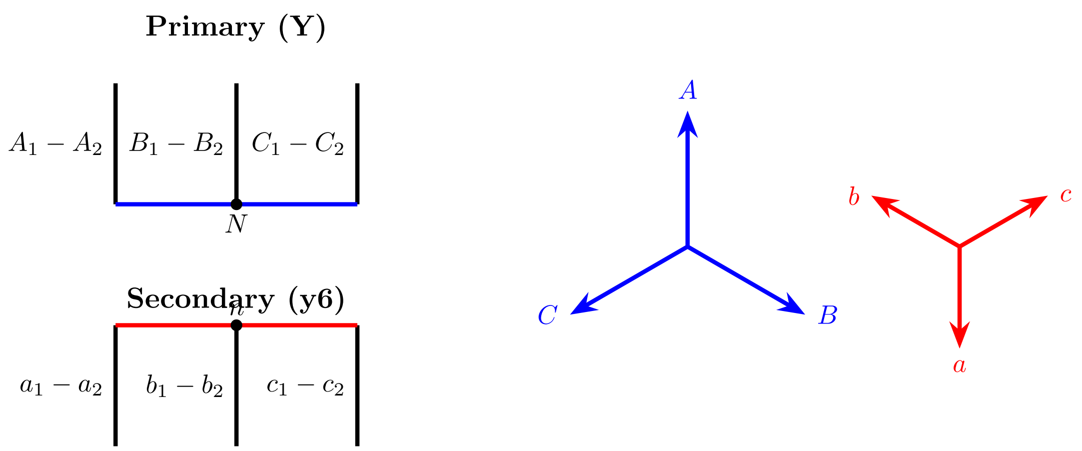

Figure 4: Yy6 Wiring and Phasor alignment. Reversing the secondary neutral point flips all vectors by 180°.

4.2 2. Vector Group: Dd6

Configuration: Delta Primary (D), Delta Secondary (d), 180° Shift.

- **Wiring Logic:** Primary connects $A_2 \to B_1$. Secondary is wired backwards: $a_1 \to b_2$, $b_1 \to c_2$, $c_1 \to a_2$. Outputs are taken from a_1, b_1, c_1.
- **Phasor Result:** The secondary delta triangle is inverted (pointing downward).

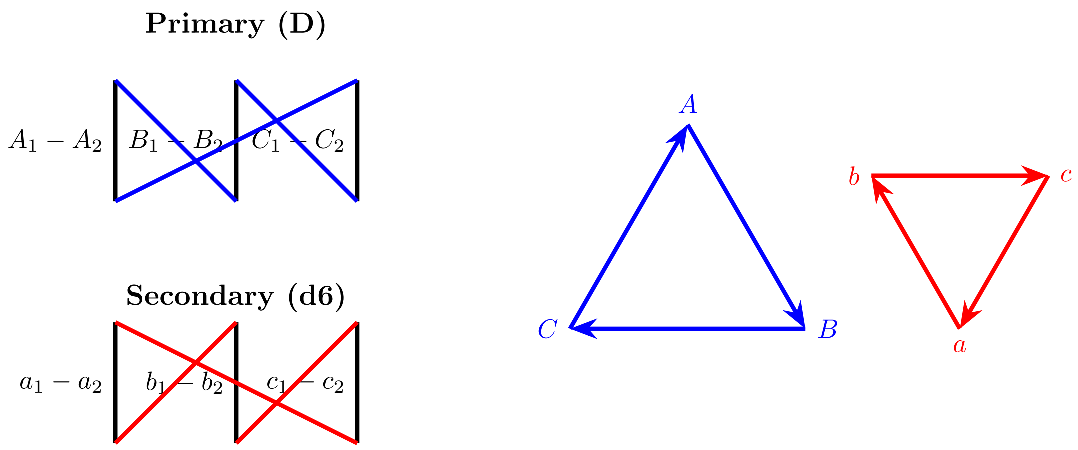

Figure 5: Dd6 Wiring and Phasor alignment. The secondary delta triangle is inverted (180°).

5 Group III: $-30°$ Phase Displacement (1 o'clock)

Cognitive Thought Process & Engineering Logic

`This is the core of most step-down utility distribution transformers. A primary Delta mixed with a secondary Star naturally yields a` $-30°$ `shift. The secondary line voltage lags the primary line voltage.`

5.1 1. Vector Group: Dy1

Configuration: Delta Primary (D), Star Secondary (y), $-30°$ Shift.

- **Wiring Logic:** Primary Delta is standard ($A_2 \to B_1$). Secondary Star is standard (a_1, b_1, c_1 neutral).
- **Phasor Result:** Secondary phase a is co-linear with primary line $A - B$. Thus, secondary line $a - b$ lags by $30°$.

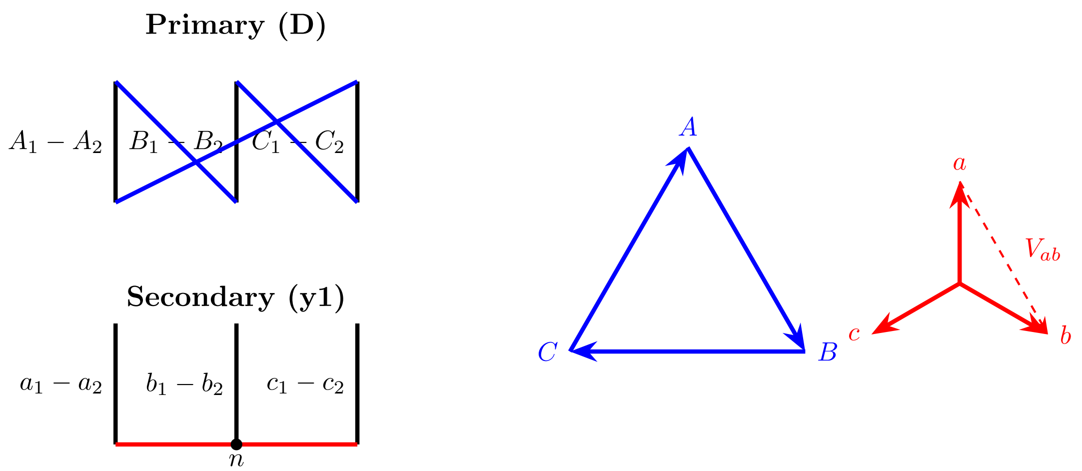

Figure 6: Dy1 Wiring and Phasor alignment. The resultant V_{ab} vector points to 1 o'clock.

5.2 2. Vector Group: Yd1

Configuration: Star Primary (Y), Delta Secondary (d), $-30°$ Shift.

- **Wiring Logic:** Primary Star (A_1, B_1, C_1 neutral). To achieve $-30°$, secondary Delta is wired $a_2 \to b_1$, $b_2 \to c_1$, $c_2 \to a_1$.

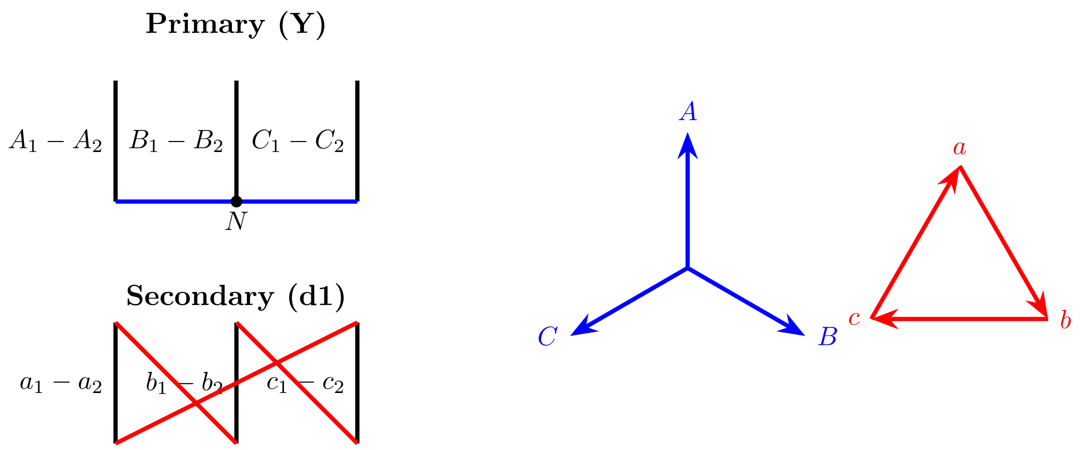

Figure 7: Yd1 Wiring and Phasor alignment. The resultant V_{ab} vector (from a to b) points to 1 o'clock.

5.3 3. Vector Group: Yz1

Configuration: Star Primary (Y), Zigzag Secondary (z), $-30°$ Shift.

- **Wiring Logic:** Zigzag splits coils across limbs. Phase a uses half of Limb 1 and half of Limb 3 in reverse.

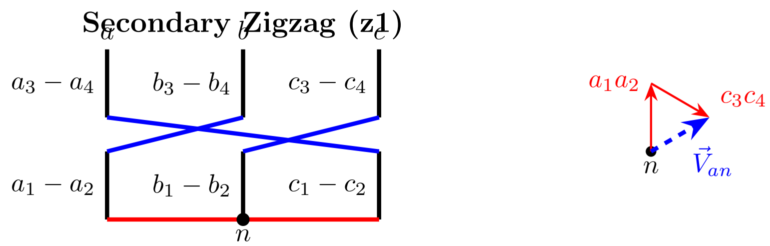

Figure 8: Yz1 Wiring. The subtraction of phase vectors across different limbs creates the $-30°$ shift.

6 Group IV: +30° Phase Displacement (11 o'clock)

Cognitive Thought Process & Engineering Logic

`If Group III (1 o'clock) lags by 30°, Group IV (11 o'clock) leads by 30°. How do we flip the shift? By crossing the Delta connections. Instead of` $A_2 \to B_1$, `we connect` $A_1 \to B_2$. `This reverses the spatial orientation of the triangle.`

6.1 1. Vector Group: Dy11

Configuration: Delta Primary (D), Star Secondary (y), +30° Shift.

- **Wiring Logic:** Primary Delta is crossed: $A_1 \to B_2$, $B_1 \to C_2$, $C_1 \to A_2$. Secondary Star is standard (a_1, b_1, c_1 neutral).

- **Phasor Result:** Secondary line $a - b$ leads primary line $A - B$ by 30°.

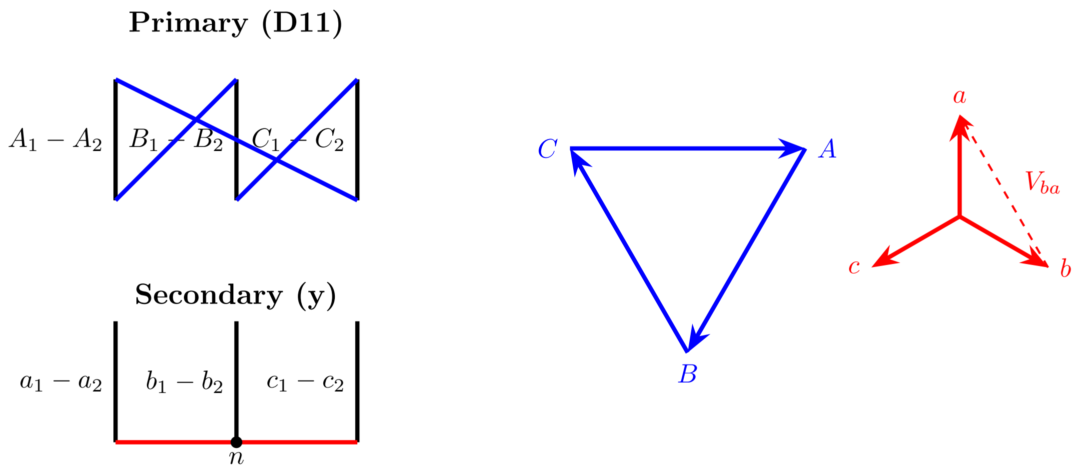

Figure 9: Dy11 Wiring and Phasor alignment. The crossed primary forces the +30° shift.

6.2 2. Vector Group: Yd11

Configuration: Star Primary (Y), Delta Secondary (d), +30° Shift.

- **Wiring Logic:** Primary Star (A_1, B_1, C_1 neutral). Secondary Delta is crossed: $a_1 \to b_2$, $b_1 \to c_2$, $c_1 \to a_2$.

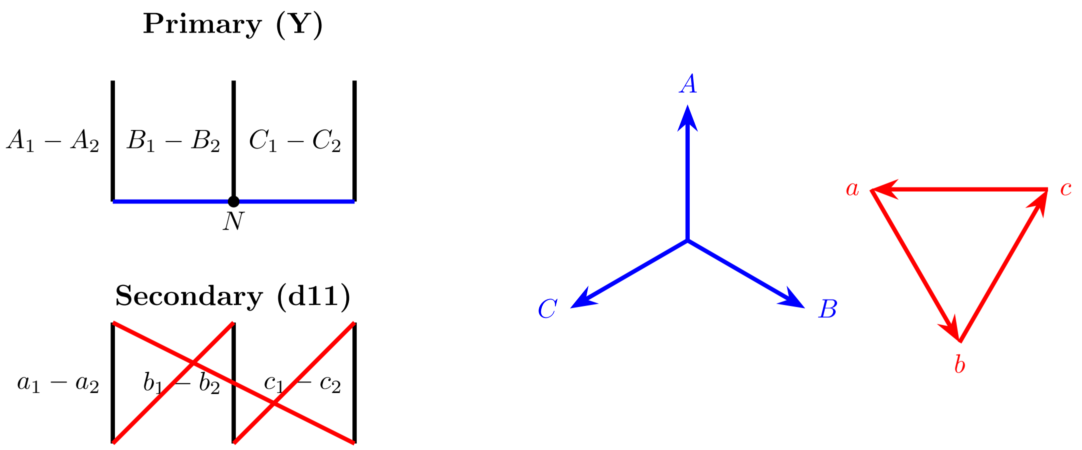

Figure 10: Yd11 Wiring and Phasor alignment. The resultant V_{ab} vector points to 11 o'clock.

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7 Parallel Operation Cross-Wiring Strategies

Cognitive Thought Process & Engineering Logic

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1. We proved earlier that connecting different vector groups (like 1
o'clock and 11 o'clock) directly to the same busbar causes a massive short
circuit.
2. However, replacing a multi-million dollar transformer just because its
clock number doesn't match is bad engineering.
3. We can manipulate the external phase sequence entering the transformer
to intentionally reverse its internal magnetic rotation, changing an 11
o'clock shift into a 1 o'clock shift!
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7.1 Matrix Reconfiguration: Paralleling Dy1 and Yd11

A $Dy1$ transformer introduces a phase displacement of $-30°$ (1 o'clock), while a $Yd11$ transformer introduces a phase displacement of $+30°$ (11 o'clock).

To bring a $Yd11$ unit into perfect alignment with an existing $Dy1$ unit, we apply a **double phase-swap** at the external bushings.

1. **Primary Swap:** Swap the incoming high-voltage grid lines for Phase B and Phase C. The transformer inputs are now ordered $A - C - B$. This reverses the internal magnetic phase rotation. The original $+30°$ lead (11 o'clock) physically mirrors across the 12 o'clock axis, becoming a $-30°$ lag (1 o'clock).

2. **Secondary Swap:** Because the magnetic field is rotating backwards, the outputs are coming out in $a - c - b$ sequence. We swap the secondary Phase B and Phase C output cables before they bolt to the LV busbar to restore standard $a - b - c$ sequence to the load.

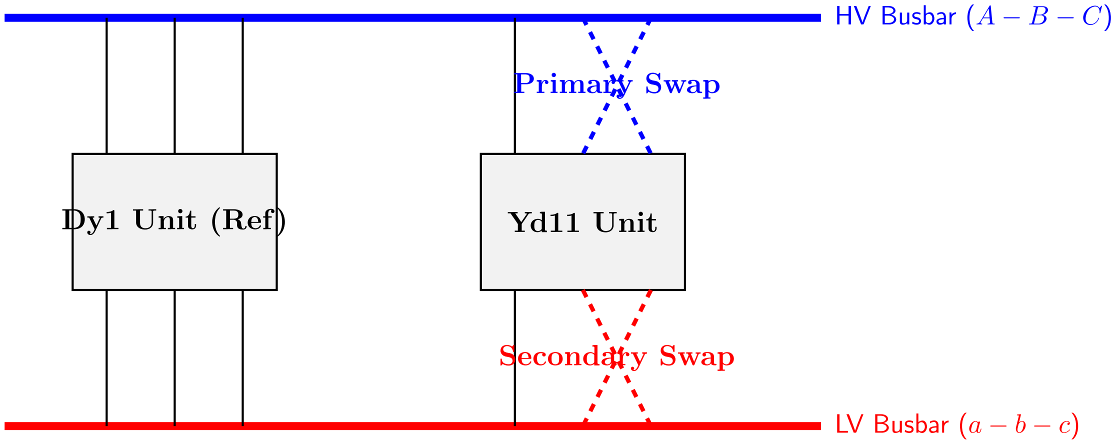

Figure 11: Double phase-swap routing to force a Yd11 unit to parallel safely with a Dy1 unit.

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8 Phase 2 Review and Self-Evaluation

Cognitive Thought Process & Engineering Logic

`Verify your engineering competency profile against the technical checklist below before advancing to Phase 3 (Rotational Magnetic Field Dynamics).`

8.1 Theoretical Competency Checklist

1. **The Displacement Proof:** Can you derive the mathematical proof showing that combining Delta and Star connections always yields an implicit $\pm 30°$ line-to-line phase shift?

2. **Clock Orientation Mapping:** If an examiner provides a $Yd11$ vector notation, can you immediately identify its phase shift angle ($+30°$ lead) and locate its positions on a standard clock face?

3. **Vector Group Schematics:** Can you confidently draw the primary and secondary winding interconnections for $Dy1$, $Yd11$, and $Yz1$ from memory?

4. **Parallel Compatibility Constraints:** Explain why a $Yy0$ unit and a $Dd6$ unit cannot be operated in parallel. Provide the exact mathematical value of the resultant potential difference ($2V$) across their terminals.

5. **External Vector Modification:** Can you reconstruct the phase-swapping configuration matrix required to run a $Dy1$ transformer in parallel with a $Yd11$ transformer?

You have successfully completed the Extended Phase 2 Masterclass. You have mastered spatial vector topology and parallel boundary configurations. You are now prepared to advance to Phase 3: Parallel Impedance Matrices.